



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Kirankumar Thanduri
14th August 2025



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- **Summary of Methodology:**

- Analyzed SpaceX launch data to identify patterns and predict launch success.
- Used REST APIs, web scraping, SQL, Folium maps, Plotly Dash, and machine learning.
- Built a dashboard and classification models to support decision-making

- **Summary of all results:**

- Payload and launch site are strong predictors of success.
- Falcon 9 boosters show high reliability.
- Dashboard and model can support launch planning and risk assessment.

Introduction

- Background: SpaceY aims to reduce launch costs and improve reliability.
- Problem Statement: Can we predict launch success based on payload, site, and booster?
- Goal: Build an interactive dashboard and predictive model using historical data of SpaceX.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - SpaceX Launch API
 - Wikipedia (scraped for booster/payload details)
- Perform data wrangling
 - • Merged API and scraped data.
 - • Cleaned nulls, standardized formats.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - How to build, tune, evaluate classification models

Data Collection

- Made a request to SpaceX API
 - Filter the dataframe to only include `Falcon 9` launches
 - Request to the SpaceX API
 - Clean the requested data
- Web scraping Falcon 9 and Falcon Heavy Launches Records from Wikipedia

Data Collection – SpaceX API

[https://github.com/kirankumarthanduri/Kirankumar-
Thanduri/blob/Capstone/jupyter-
labs-spacex-data-collection-api.ipynb](https://github.com/kirankumarthanduri/Kirankumar-
Thanduri/blob/Capstone/jupyter-
labs-spacex-data-collection-api.ipynb)

-
- ```
graph TD; A[• Import Libraries and Define Auxiliary Functions] --> B[• Requesting and parsing the SpaceX launch data using the GET request]; B --> C[• Filter the dataframe to only include 'Falcon 9' launches]; C --> D[• Dealing with Missing Values];
```
- Import Libraries and Define Auxiliary Functions
  - Requesting and parsing the SpaceX launch data using the GET request
  - Filter the dataframe to only include `Falcon 9` launches
  - Dealing with Missing Values



# Data Collection - Scraping

---

[https://github.com/kirankumarthanduri/Kirankumar-  
Thanduri/blob/Capstone/labs-  
jupyter-spacex-  
Data%20wrangling.ipynb](https://github.com/kirankumarthanduri/Kirankumar-<br/>Thanduri/blob/Capstone/labs-<br/>jupyter-spacex-<br/>Data%20wrangling.ipynb)

- Web scraping Falcon 9 and Falcon Heavy Launches Records from Wikipedia
- ↓
- Request the Falcon9 Launch Wiki page from its URL
- ↓
- Extract all column/variable names from the HTML table header
- ↓
- Create a data frame by parsing the launch HTML tables

# Data Wrangling

---

[https://github.com/kirankumarthanduri/Kirankumar-  
Thanduri/blob/Capstone/labs-  
jupyter-spacex-  
Data%20wrangling.ipynb](https://github.com/kirankumarthanduri/Kirankumar-<br/>Thanduri/blob/Capstone/labs-<br/>jupyter-spacex-<br/>Data%20wrangling.ipynb)

## Space X Falcon 9 First Stage Landing Prediction - Data Analysis

- 
- ```
graph TD; A[Space X Falcon 9 First Stage Landing Prediction - Data Analysis] --> B[Calculate the number of launches on each site]; B --> C[Calculate the number and occurrence of each orbit]; C --> D[Calculate the number and occurrence of mission outcome of the orbits]; D --> E[Create a landing outcome label from Outcome column];
```
- Calculate the number of launches on each site
 - Calculate the number and occurrence of each orbit
 - Calculate the number and occurrence of mission outcome of the orbits
 - Create a landing outcome label from Outcome column

EDA with Data Visualization

[https://github.com/kirankumarthanduri/Kirankumar-
Thanduri/blob/Capstone/edadataviz.i
pynb](https://github.com/kirankumarthanduri/Kirankumar-
Thanduri/blob/Capstone/edadataviz.ipynb)

Few of the EDA and Data Visualization steps performed:

- Read the SpaceX dataset into a Pandas dataframe and print its summary
- Visualize the relationship between Flight Number and Launch Site
- Visualize the relationship between Payload Mass and Launch Site
- Visualize the relationship between success rate of each orbit type
- Visualize the relationship between FlightNumber and Orbit type
- Visualize the relationship between Payload Mass and Orbit type
- Visualize the launch success yearly trend
- Create dummy variables to categorical columns
- Cast all numeric columns to float64

EDA with SQL

[https://github.com/kirankumarthanduri/Kirankumar-
Thanduri/blob/Capstone/jupyter-
labs-eda-sql-coursera_sqlite.ipynb](https://github.com/kirankumarthanduri/Kirankumar-
Thanduri/blob/Capstone/jupyter-
labs-eda-sql-coursera_sqlite.ipynb)

- Display the names of the unique launch sites in the space mission
- Display 5 records where launch sites begin with the string 'CCA'
- Display the total payload mass carried by boosters launched by NASA (CRS)
- Display average payload mass carried by booster version F9 v1.1
- List the date when the first successful landing outcome in ground pad was achieved.
- List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000
- List the total number of successful and failure mission outcomes
- List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function
- List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.
- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

Build an Interactive Map with Folium

[https://github.com/kirankumarthanduri/Kirankumar-
Thanduri/blob/Capstone/lab_jupyter_launch_site_location.ipynb](https://github.com/kirankumarthanduri/Kirankumar-
Thanduri/blob/Capstone/lab_jupyter_launch_site_location.ipynb)

Below are the Maps and Markers created using Folium :

- Mark all launch sites on a map
 - Add each site's location on a map using site's latitude and longitude coordinates
 - folium.Circle and folium.Marker are used to highlight the launch areas.
- Mark the success/failed launches for each site on the map
- Calculate the distances between a launch site to its proximities

Build a Dashboard with Plotly Dash

Predictive Analysis (Classification)

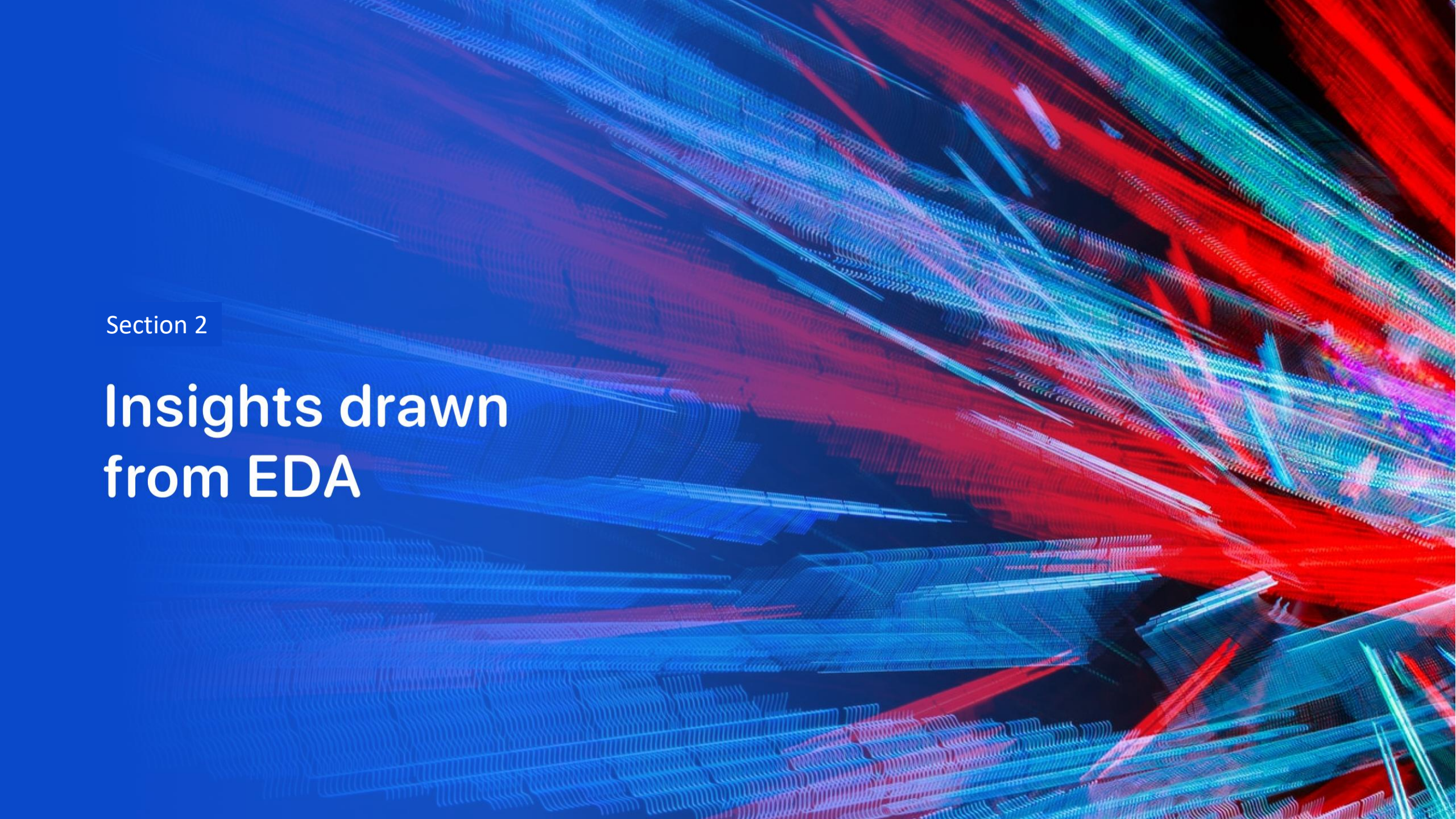
[https://github.com/kirankumarthanduri/Kirankumar-
Thanduri/blob/Capstone/SpaceX
Machine%20Learning%20Prediction
Part 5.ipynb](https://github.com/kirankumarthanduri/Kirankumar-
Thanduri/blob/Capstone/SpaceX
Machine%20Learning%20Prediction
Part%205.ipynb)

- Perform exploratory Data Analysis and determine Training Labels
 - Create a NumPy array from the column Class in data, by applying the method to `_numpy()` then assign it to the variable Y, make sure the output is a Pandas series (only one bracket `df['name of column']`).
 - Standardize the data in X then reassign it to the variable X using the transform
 - Use the function `train_test_split` to split the data X and Y into training and test data. Set the parameter `test_size` to 0.2 and `random_state` to 2. The training data and test data should be assigned to the following labels.
 - Create a logistic regression object then create a `GridSearchCV` object `logreg_cv` with `cv = 10`. Fit the object to find the best parameters from the dictionary parameters.
 - Calculate the accuracy on the test data using the different methods

Results

- Logistic Regression-accuracy : 0.8464285714285713
- SVM: accuracy-accuracy : 0.8482142857142856
- Decision Tree-accuracy : 0.875
- KNN: accuracy-accuracy : 0.8482142857142858

- Model_accuracies
 - Best Model: Decision Tree
 - Best Accuracy on test data: 0.94

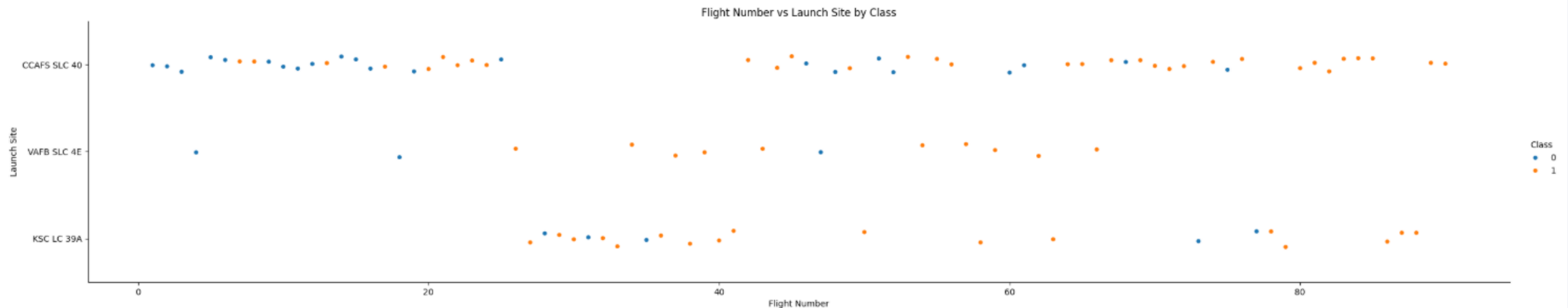
The background of the slide is an abstract composition. It features a dark blue field on the left side, which transitions into a complex pattern of diagonal streaks and lines in shades of blue, red, and cyan on the right. These streaks have a textured, almost woven appearance, suggesting a digital or data-driven theme. The overall effect is dynamic and modern.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

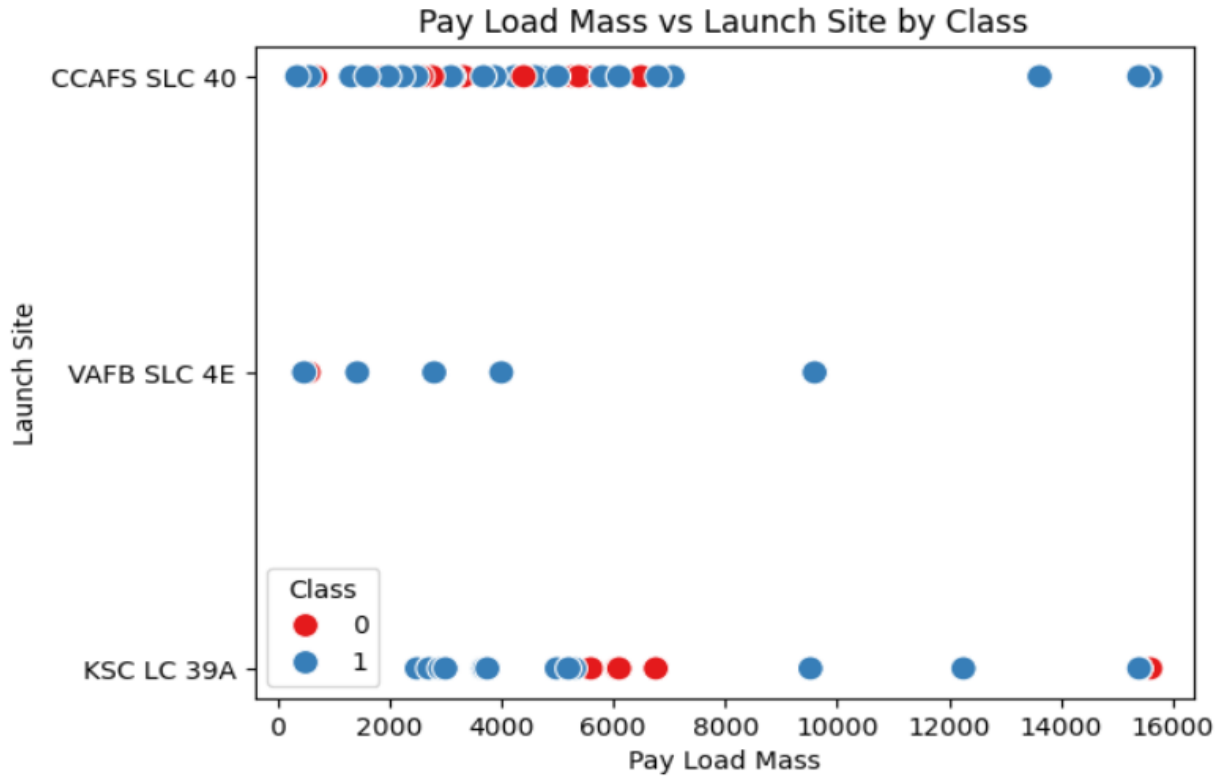
```
[7]: # Plot a scatter point chart with x axis to be Flight Number and y axis to be the launch site, and hue to be the class value
sns.catplot(x="FlightNumber", y="LaunchSite", hue="Class", data=df, aspect=5)
plt.title("Flight Number vs Launch Site by Class")
plt.xlabel("Flight Number")
plt.ylabel("Launch Site")
plt.show()
```



Now try to explain the patterns you found in the Flight Number vs. Launch Site scatter point plots.

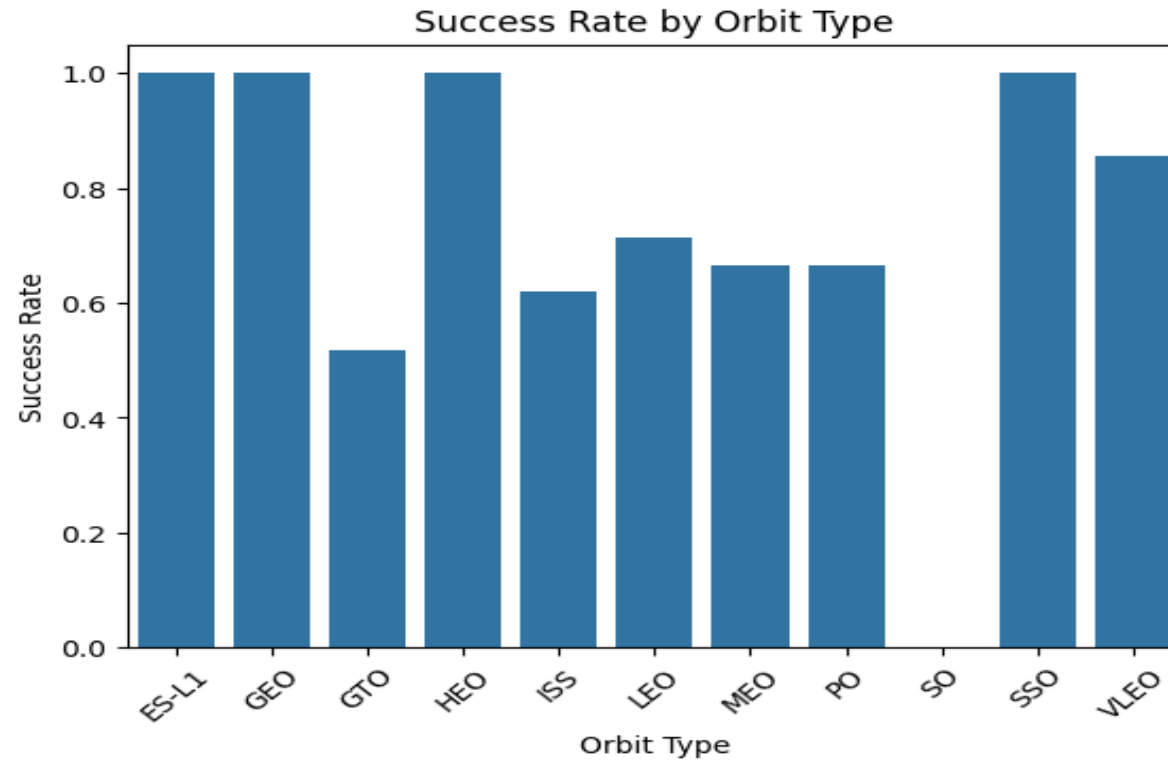
- Temporal Improvement: Success rate improves with flight number, reflecting SpaceX's learning curve and technological refinement.
- Site Reliability: KSC LC 39A emerges as the most reliable site in later missions.
- Early Risk: Initial launches (low flight numbers) had higher failure rates, regardless of site.

Payload vs. Launch Site



Now if you observe Payload Mass Vs. Launch Site scatter point chart you will find for the VAFB-SLC launchsite there are no rockets launched for heavypayload mass(greater than 10000).

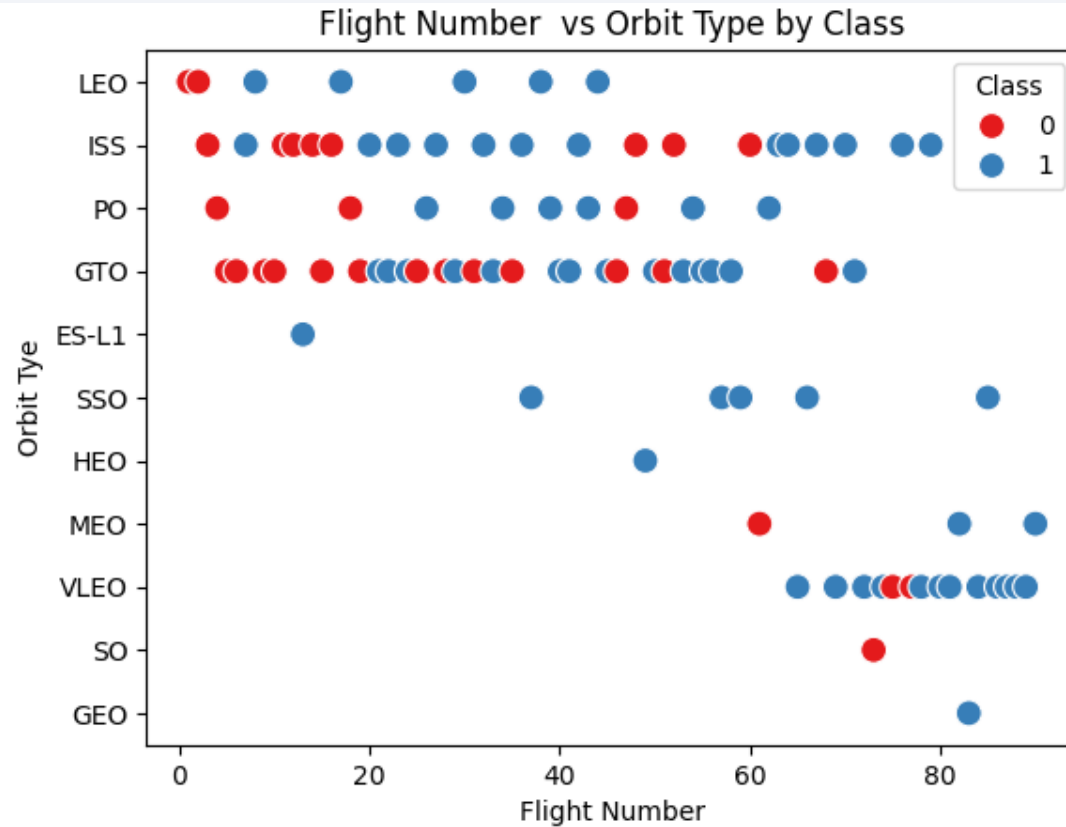
Success Rate vs. Orbit Type



Analyze the plotted bar chart to identify which orbits have the highest success rates.

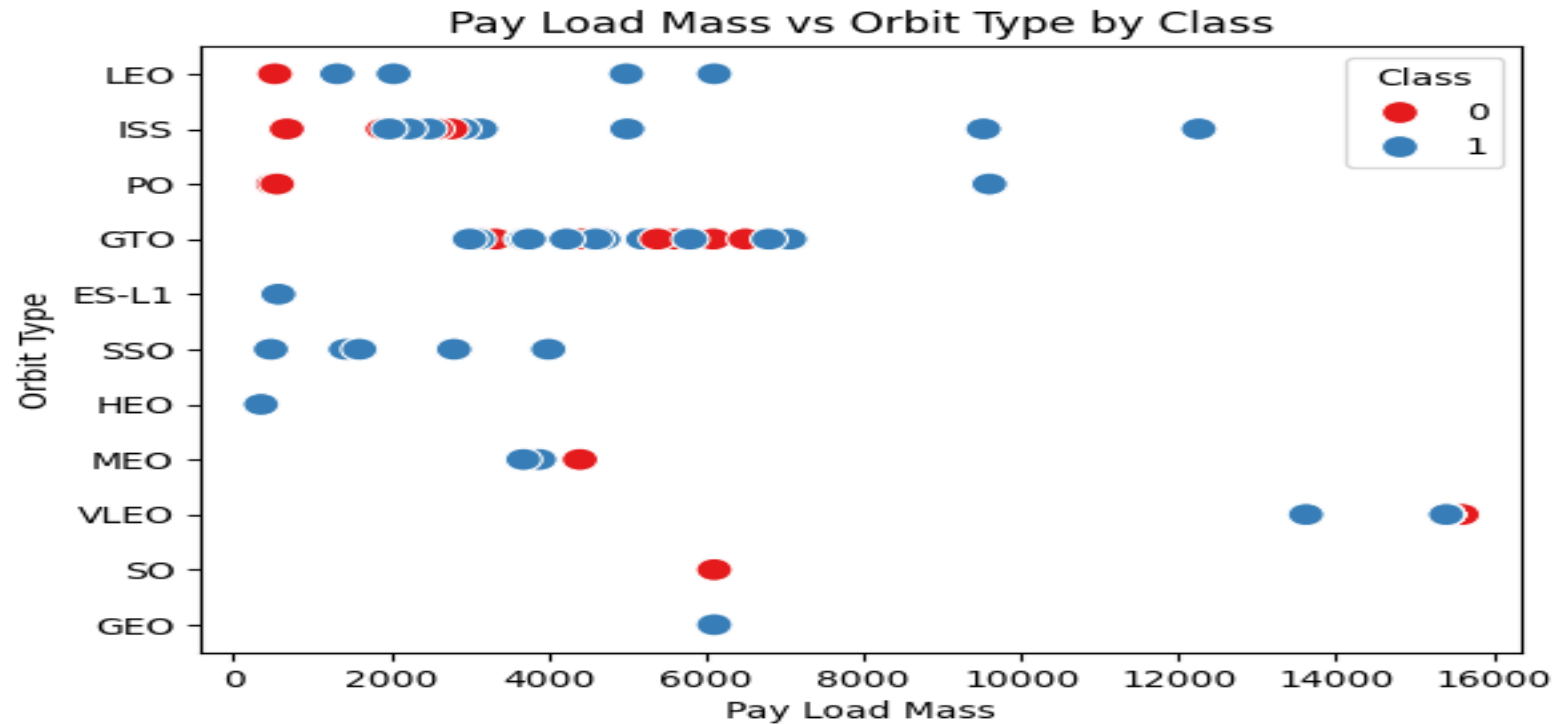
- High success rates in ES-L1, GEO, HEO, and SSO suggest strong mission reliability for scientific and commercial payloads.
- Lower success in GTO may reflect the technical challenges of transfer orbits and multi-payload deployments.

Flight Number vs. Orbit Type



You can observe that in the LEO orbit, success seems to be related to the number of flights. Conversely, in the GTO orbit, there appears to be no relationship between flight number and success.

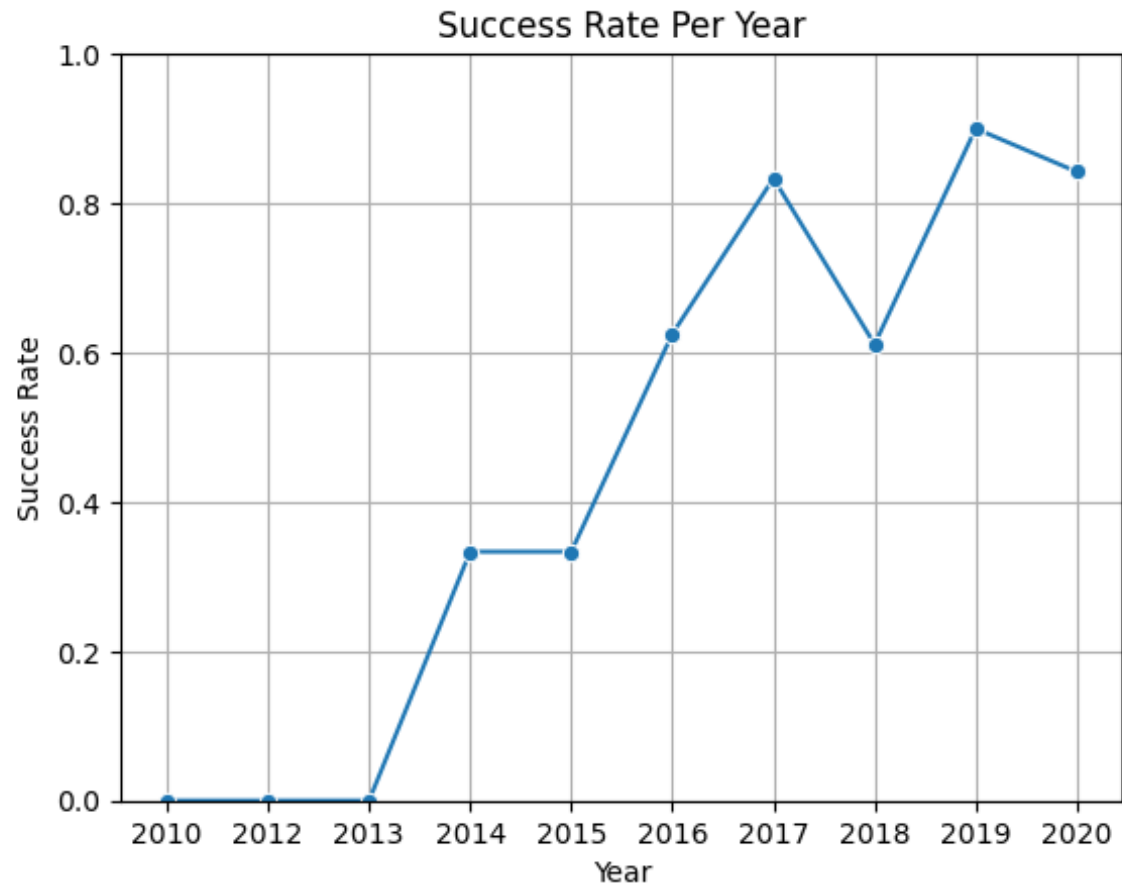
Payload vs. Orbit Type



With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS.

However, for GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present.

Launch Success Yearly Trend



you can observe that the success rate since 2013 kept increasing till 2020

All Launch Site Names

- Below are the 4 unique Launch sites which have been pulled using SQL query:

- CCAFS LC-40
- VAFB SLC-4E
- KSC LC-39A
- CCAFS SLC-40

Task 1

Display the names of the unique launch sites in the space mission

```
%sql SELECT DISTINCT "launch_site" FROM SPACE_TABLE;
```

```
* sqlite:///my_data1.db
```

Done.

Launch_Site

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

Launch Site Names Begin with 'CCA'

Display 5 records where launch sites begin with the string 'CCA'.

```
[24]: %sql SELECT * FROM SPACEXTABLE where launch_site Like "CCA%" Limit 5;
```

```
* sqlite:///my_data1.db
```

Done.

```
[24]:
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

```
[16]: %sql SELECT Sum(PAYLOAD_MASS__KG_) as Total_PayLoad_NASA FROM SPACEXTABLE where Customer Like "NASA (CRS)";
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[16]: Total_PayLoad_NASA
```

```
45596
```

- Total Payload Mass in KGs at NASA (CRS) is calculated using above query which results in 4559 KG

Average Payload Mass by F9 v1.1

Task 4

Display average payload mass carried by booster version F9 v1.1

```
: %sql SELECT AVG(PAYLOAD_MASS__KG_) as Avg_PayLoad_F9V1 FROM SPACEXTABLE where Booster_Version Like "F9 v1.1%";  
* sqlite:///my_data1.db  
Done.  
: Avg_PayLoad_F9V1  
-----  
2534.6666666666665
```

- Average Payload Mass in KGs at NASA (CRS) is calculated using above query which results in 2534.67 KG

First Successful Ground Landing Date

Python3 Jupyter Notebook

```
[29]: %sql SELECT min(Date) as Succussful_Ground_Landing FROM SPACEXTABLE where Landing_Outcome = "Success (ground pad)";
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[29]: Succussful_Ground_Landing
```

```
2015-12-22
```

- First successful landing outcome on ground pad was calculated using above query which resulted in 22nd December 2015

Successful Drone Ship Landing with Payload between 4000 and 6000

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
30]: %sql SELECT Booster_Version FROM SPACEXTABLE where Landing_Outcome = "Success (drone ship)" and PAYLOAD_MASS__KG_ between 4000 and 6000;  
* sqlite:///my_data1.db  
Done.
```

```
30]: Booster_Version
```

```
F9 FT B1022
```

```
F9 FT B1026
```

```
F9 FT B1021.2
```

```
F9 FT B1031.2
```

Names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000 are calculated using above query, they are:

- F9 FT B1022
- F9 FT B1026
- F9 FT B1021.2
- F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

Task 7

List the total number of successful and failure mission outcomes

```
[33]: %sql SELECT Count(Landing_Outcome) as Successful_Failure_Mission_Outcomes FROM SPACEXTABLE where Landing_Outcome Like "Success%" or Landing_Outcome Li
* sqlite:///my_data1.db
Done.
[33]: Successful_Failure_Mission_Outcomes
```

71

- Total number of successful and failure mission outcomes are calculated as: 71

Boosters Carried Maximum Payload

```
[35]: %sql SELECT Booster_Version FROM SPACEXTABLE where PAYLOAD_MASS_KG_ = (select max(PAYLOAD_MASS_KG_) from SPACEXTABLE);
* sqlite:///my_data1.db
Done.
[35]: Booster_Version
```

F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

- The names of the boosters which carried the maximum payload mass are:
- F9 B5 B1048.4, F9 B5 B1049.4, F9 B5 B1051.3, F9 B5 B1056.4, F9 B5 B1048.5, F9 B5 B1051.4, F9 B5 B1049.5, F9 B5 B1060.2, F9 B5 B1058.3, F9 B5 B1051.6, F9 B5 B1060.3, F9 B5 B1049.7

2015 Launch Records

```
[40]: %sql SELECT Date, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTABLE where Date like "%2015%" and Landing_Outcome ="Failure (drone ship)";
* sqlite:///my_data1.db
Done.
```

```
[40]:
```

Date	Landing_Outcome	Booster_Version	Launch_Site
2015-01-10	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
2015-04-14	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

- Failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

<u>Date</u>	<u>Landing_Outcome</u>	<u>Booster_Version</u>	<u>Launch_Site</u>
2015-01-10	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
2015-04-14	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
Task 10
Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

[46]: %sql SELECT Landing_Outcome, Count(*) as Outcome FROM SPACEXTABLE where Date BETWEEN '2010-06-04' and '2017-03-20' GROUP BY Landing_Outcome order by Outcome desc

* sqlite:///my_data1.db
Done.

[46]:
```

Landing_Outcome	Outcome
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

- Ranked count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20 are:

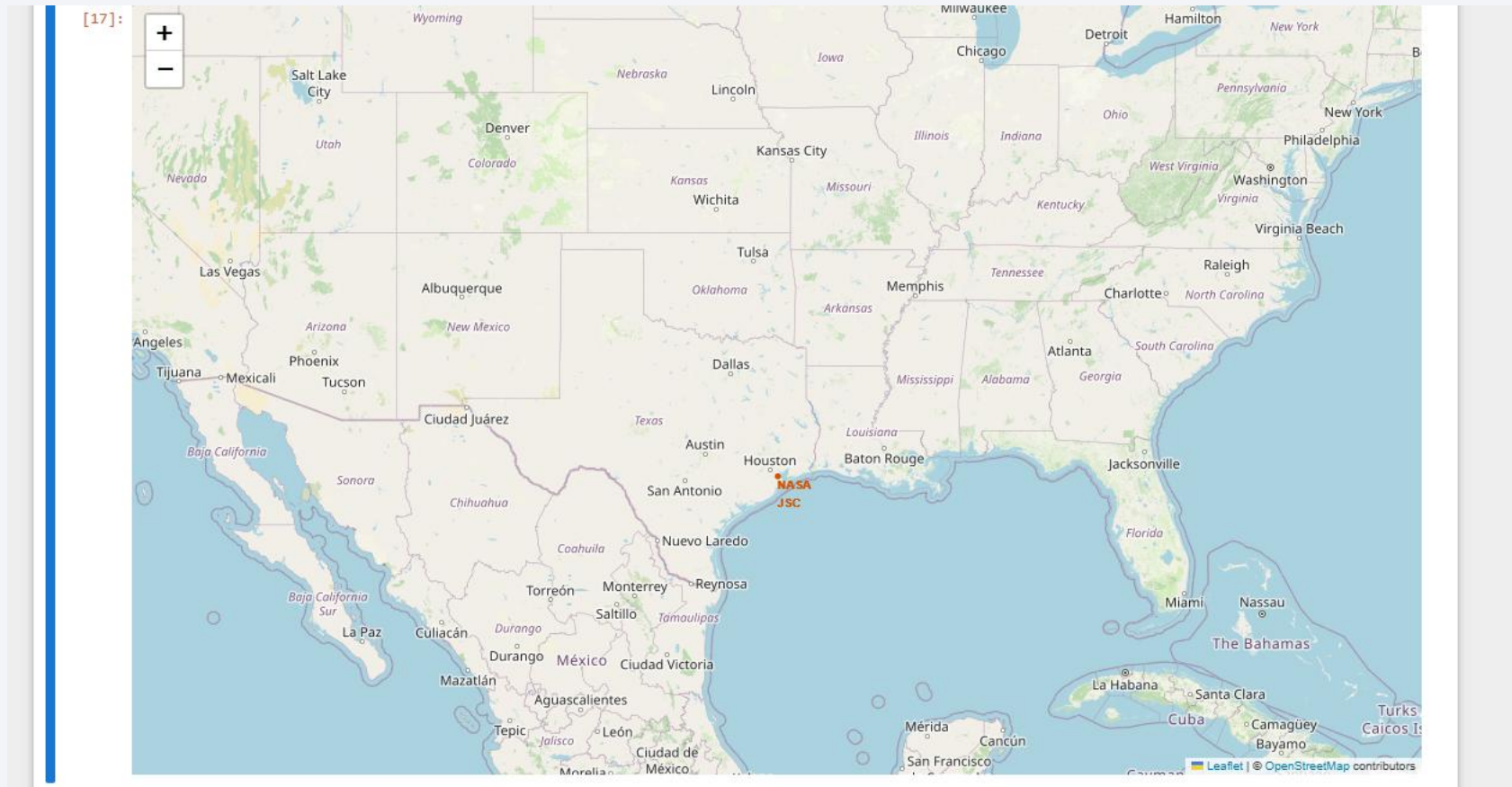
<u>Landing_Outcome</u>	<u>Outcome</u>
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky and a view of the Earth's surface, which is covered in a dense network of city lights and clouds. The lights are concentrated in the lower right portion of the image, while the upper left shows a clear blue sky.

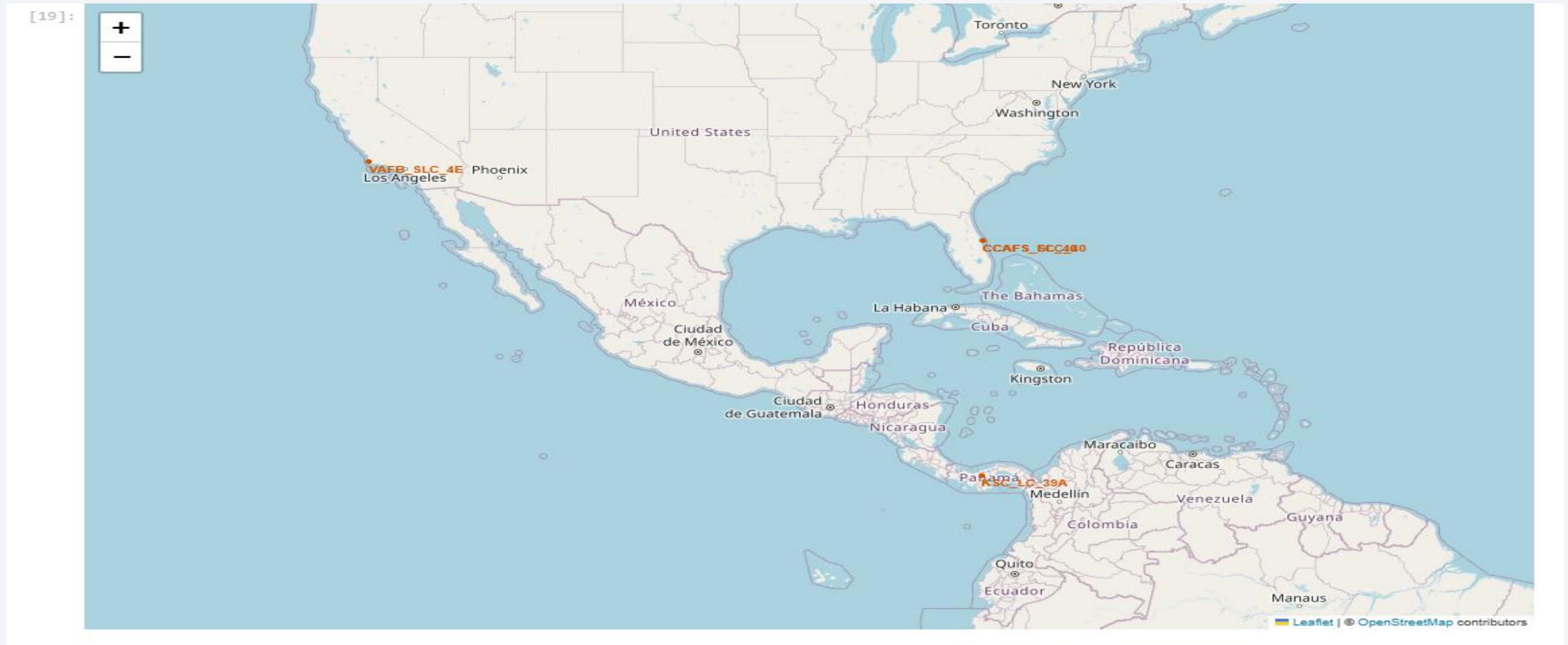
Section 3

Launch Sites Proximities Analysis

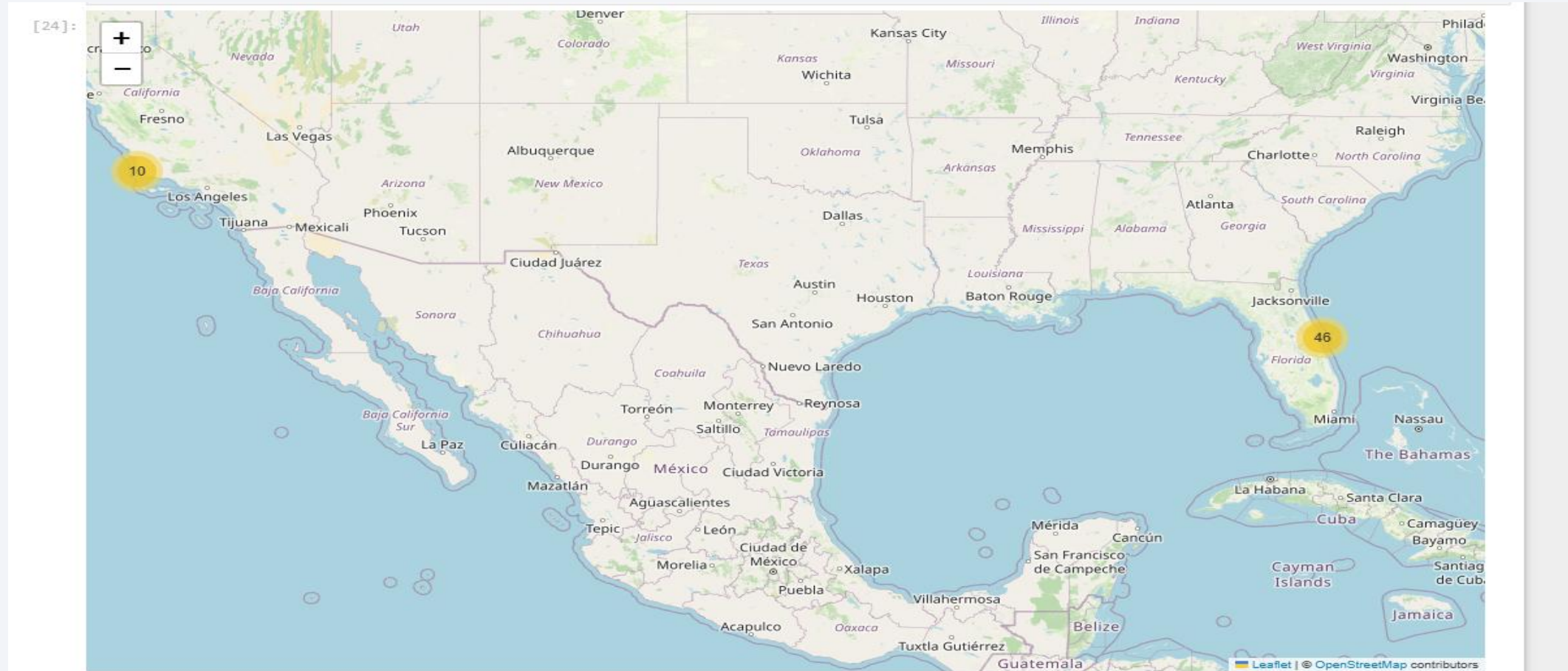
NASA Johnson Space Center at Houston, Texas.

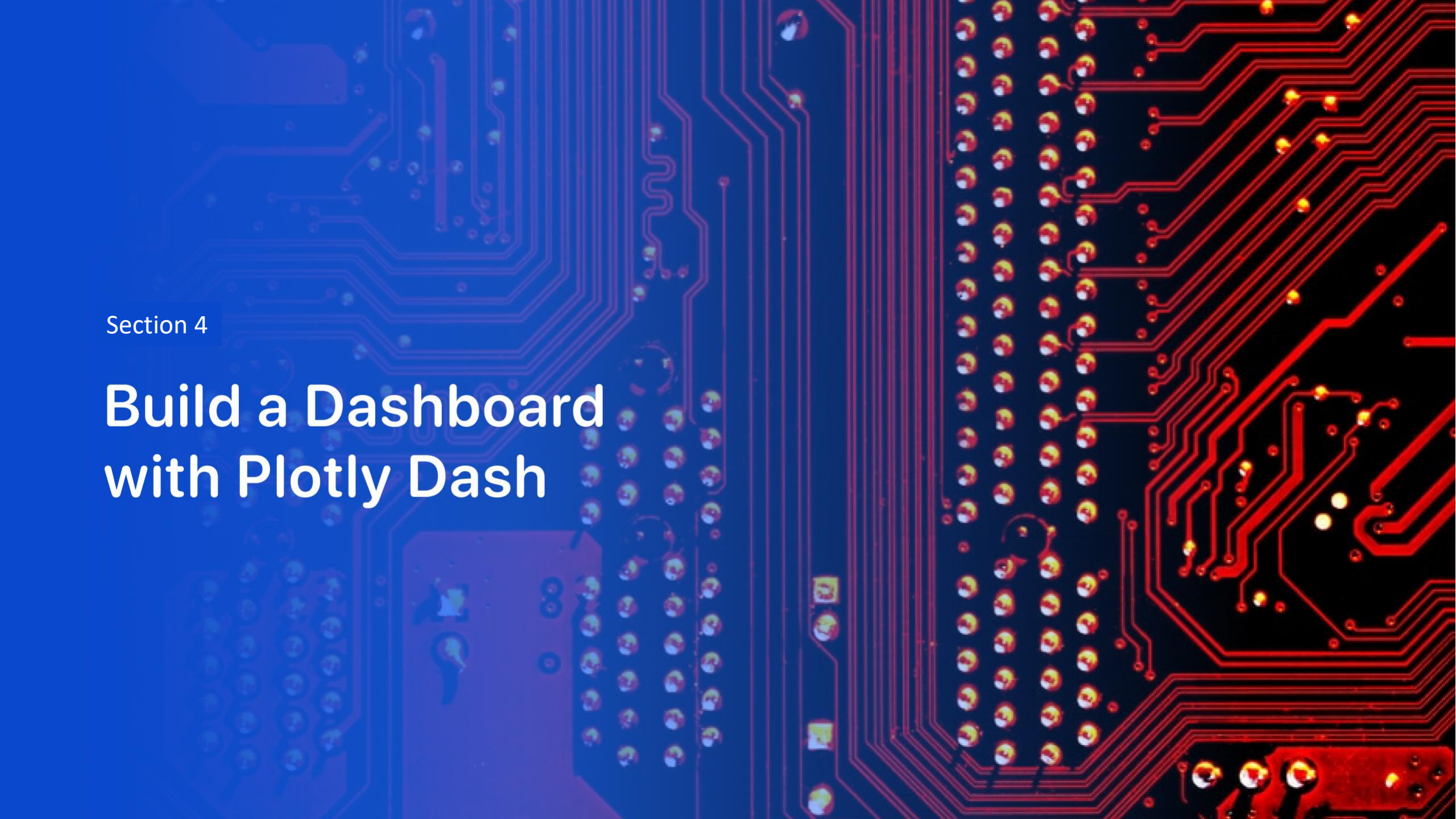


All Launch Sites



Success/failed launches for each site





Section 4

Build a Dashboard with Plotly Dash

<Dashboard Screenshot 1>

- Replace <Dashboard screenshot 1> title with an appropriate title
- Show the screenshot of launch success count for all sites, in a piechart
- Explain the important elements and findings on the screenshot

<Dashboard Screenshot 2>

- Replace <Dashboard screenshot 2> title with an appropriate title
- Show the screenshot of the piechart for the launch site with highest launch success ratio
- Explain the important elements and findings on the screenshot

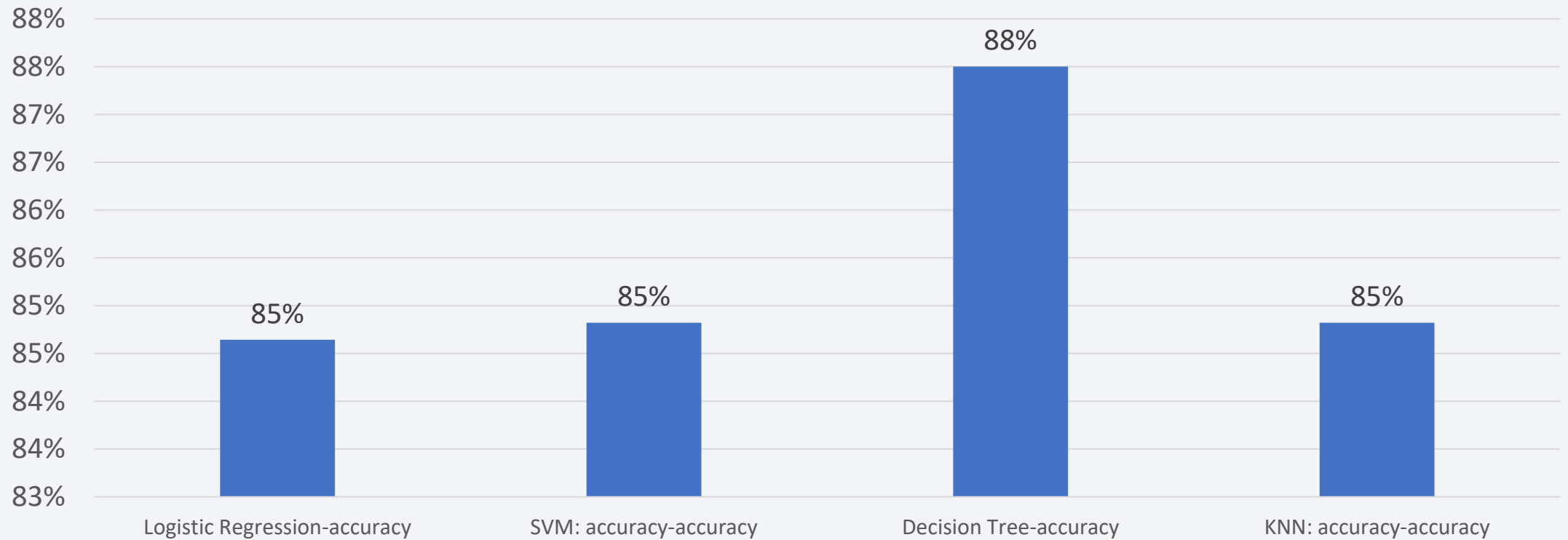
<Dashboard Screenshot 3>

- Replace <Dashboard screenshot 3> title with an appropriate title
- Show screenshots of Payload vs. Launch Outcome scatter plot for all sites, with different payload selected in the range slider
- Explain the important elements and findings on the screenshot, such as which payload range or booster version have the largest success rate, etc.

Section 5

Predictive Analysis (Classification)

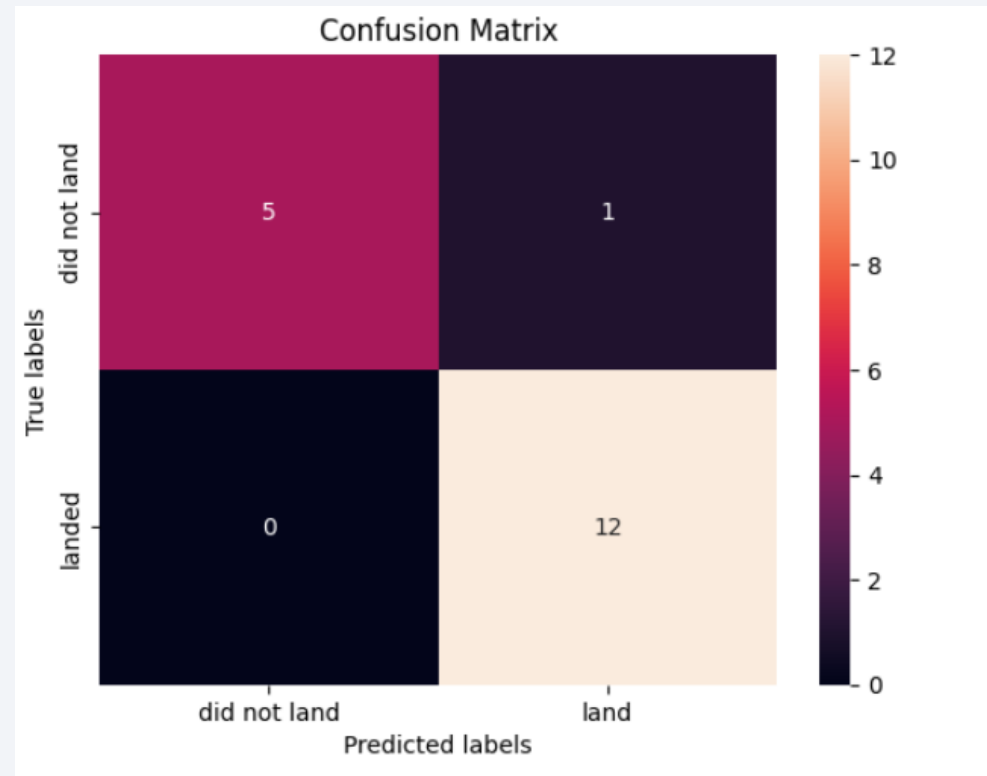
Classification Accuracy



Best Model: **Decision Tree**

Confusion Matrix

- This model is highly reliable for predicting successful landings, which is critical for SpaceX's operational planning. Its ability to avoid false negatives makes it especially valuable in risk-sensitive environments.
- Would you like me to help write this up as a slide or speaker note for your presentation? I can also compare this model's performance with others you tested.



Conclusions

- The model is very accurate (94%)
 - It correctly predicted most of the rocket landings and non-landings.
- It didn't miss any actual landings
 - Every rocket that landed was correctly identified by the model.
- Only one wrong prediction
 - Just one case where the model said "landed" but it actually didn't.
- Works well for both types of outcomes
 - It handles both "landed" and "did not land" predictions without bias.
- Can be used for future launches
 - The approach can be applied to new data to help predict future rocket landings.

Appendix

- <https://github.com/kirankumarthanduri/Kirankumar-Thanduri/tree/Capstone>

All the Jupyter Notebooks and other files available in above path.

Thank you!

